The Upper limb

The upper limb consists of 30 separate bones and is specialized for mobility. The skeleton of the arm, the region between the shoulder and elbow is the **humerus**. The head of humerus articulates with the **glenoid cavity** of the scapula and forms the shoulder joint. The distal end of humerus articulates with the two forearm bones the **radius** and **ulna**. The forearm is the region between the elbow and the wrist. **Olecranon process** is situated at the upper end of the ulna which forms the pointed portion of the elbow.The hand consists of **carpals, metacarpals** and **phalanges.**

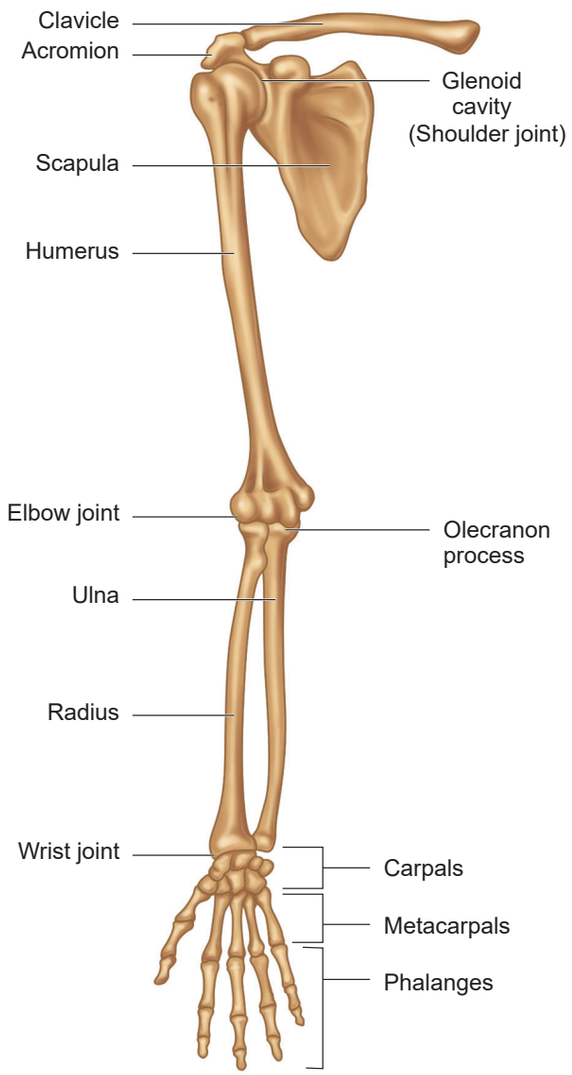


Figure 9. 8 Pectoral girdle with upper limb

Carpals,the wrist bones, 8 in number are arranged in two rows of four each. The anterior surface of the wrist has tunnel-like appearance, due to the arrangement of carpals with the ligaments. This tunnel is termed as **carpal tunnel**.

CTS-(Carpal Tunnel Syndrome) – The narrow passage (tunnel) bounded by bones and ligaments in the wrist gets narrowed and pinches the median nerve. This syndrome is mostly seen among the clerks, software professionals and pregnant women and people who constantly play or text in mobile phones.

Metacarpals,the palm bones are 5 in number and **phalanges** the digits bones are 14 in number.

(b) Pelvic Girdle

The pelvic girdle is a heavy structure specialized for weight bearing. It is composed of two hip bones called coxal bones that secure the lower limbs to the axial skeleton (Figure 9.9). Together, with the sacrum and coccyx, the hip bones form the basin-like bony pelvis.

Each coxal bone consists of three fused bones, **ilium, ischium** and **pubis**. At the point of fusion of ilium, ischium, and pubis a deep hemispherical socket called the acetabulum is present on the lateral surface of the pelvis. It receives the head of the femur or thigh bone at the hip joint and helps in the articulation of the femur. Ventrally the two halves of the pelvic girdle meet and form the **pubic symphysis** containing fibrous cartilage.

The **ilium** is the superior flaring portion of the hip bone. Each ilium forms a secure joint with the sacrum posteriorly.

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